

Key Concepts Gallery

Sagan Space Workshop study guide — beginner diagrams from basic to advanced

These are the key-concept drawings made for the notes. Each page has a short explanation and the matching diagram.

Contents

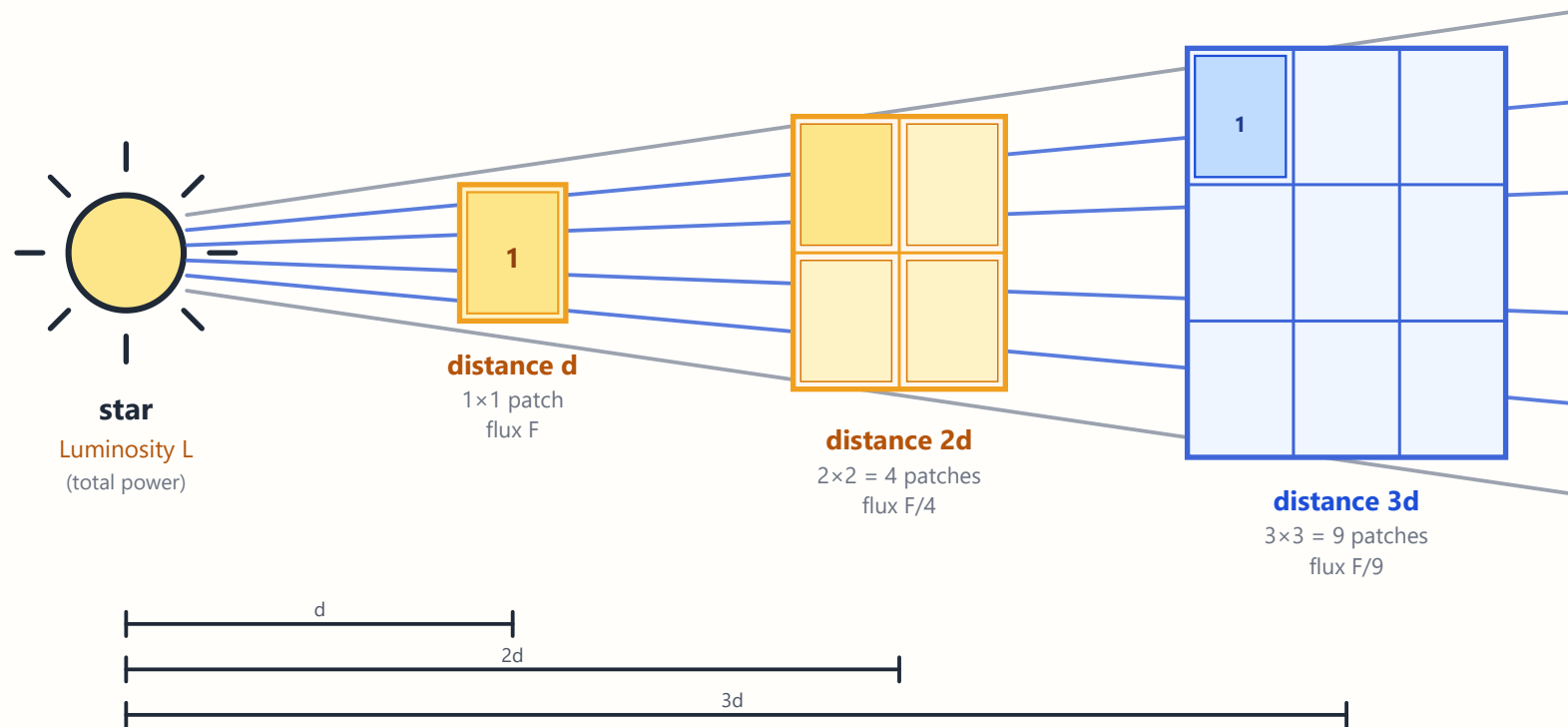
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|------------------------------|---------------------|
| 0. Light and distance | 5a. Transits |
| 1. Roman survey | 5b. Transit vetting |
| 2. Planet formation | 6. Galaxy and dust |
| 3a. Microlensing | 7. Demographics |
| 3b. Light-curve measurements | 8a. Pipelines |
| 4a. Physical clues | 8b. Host stars |
| 4b. Model ladder | 9. Follow-up |
| 4c. Model fitting loop | 10. Hands-on loop |
| 4d. Survey yield | |

Level 0 — Foundations

Light and distance

Light leaves a star with total power called luminosity (L). As it travels, that same light spreads over a larger and larger sphere. The classic picture uses equal-size patches at distances d , $2d$, and $3d$: the light that fills 1 patch at d fills 4 patches at $2d$ and 9 at $3d$. Apparent brightness (flux) therefore falls as $1/d^2$ — which is why magnitudes, distance moduli, and extinction corrections are all linked.

Light spreads out as it travels



Flux $F = L / (4\pi d^2)$ → twice as far means four times fainter

Same total light, larger area → lower brightness. That links magnitudes, distance, and extinction.

Level 1 — Roman and the GBTDS

Roman watches the same stars again and again

Roman stares at fields near the Galactic bulge and re-images them on a short cadence. Any star that brightens (microlensing) or dims (transit) leaves a time series. One camera, two planet-finding methods.

Roman watches the same stars again and again



One survey — two ways to find planets

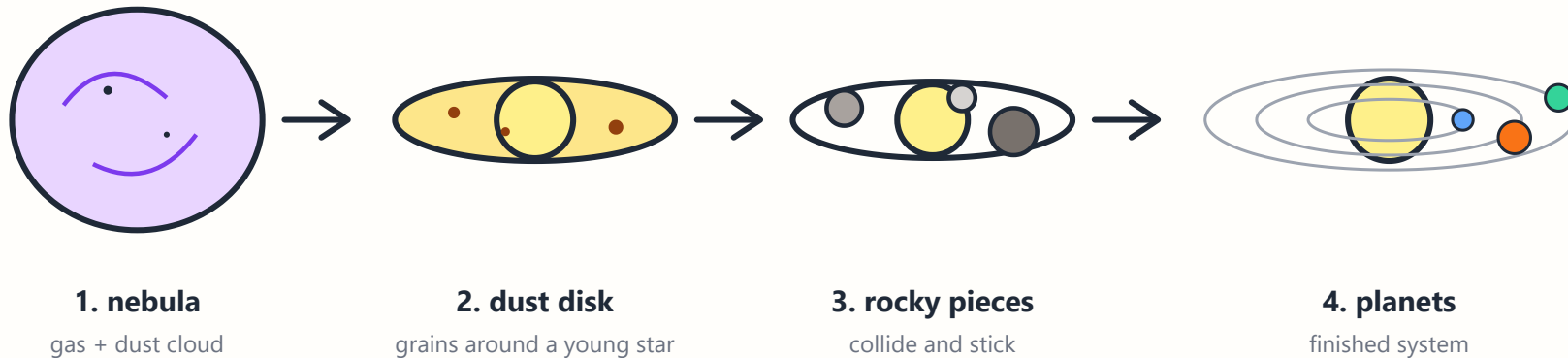
Repeated imaging turns a camera into a time-domain survey.

Level 2 — Planet formation

Planets grow inside a disk

Dust grains collide and stick into rocky pieces, and those pieces grow into young planets inside a rotating disk around a new star. The snow line, disk metallicity, and planet populations are all statements about steps in this chain.

Planets grow inside a disk



Small pieces collide, stick, and slowly build worlds.

The snow line and disk metallicity shape final planet populations.

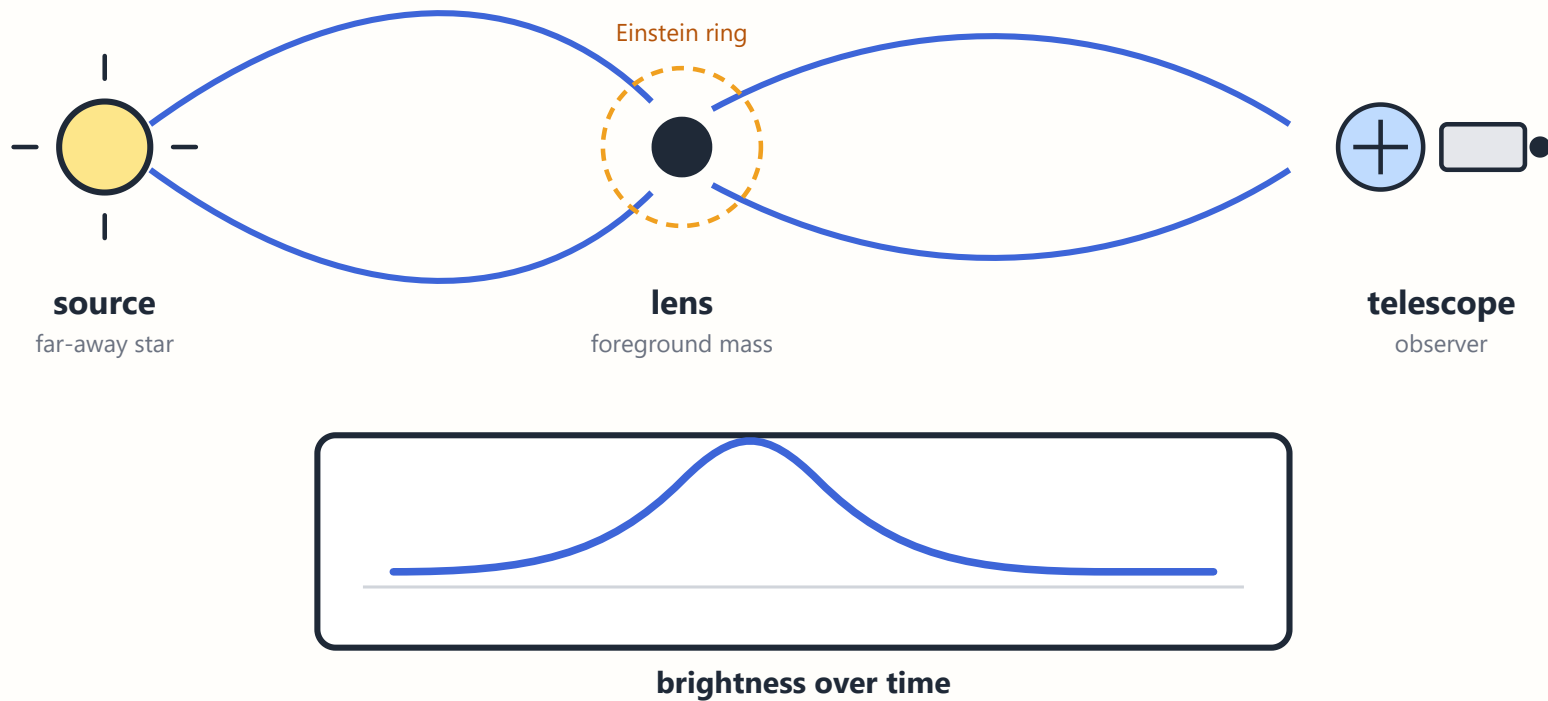
That sequence is step one for understanding Roman yields.

Level 3 — Microlensing fundamentals

Gravity bends light

A foreground lens passes almost exactly in front of a far-away source. Gravity bends and focuses the source light, so the source looks temporarily brighter. The lens does not need to shine — dark objects can still be weighed.

Microlensing = gravity bends light



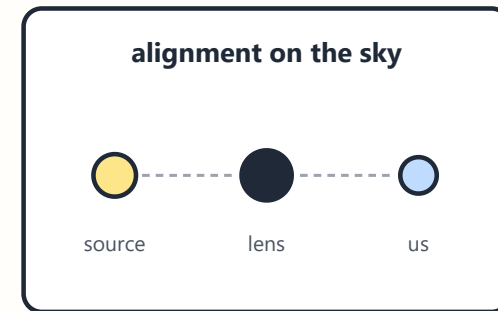
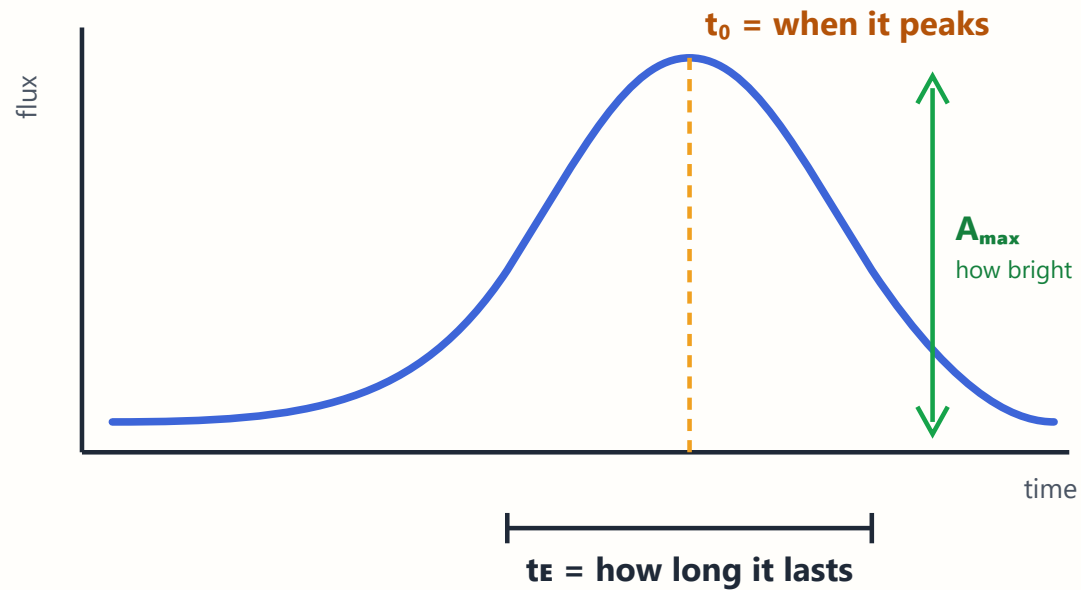
No light from the lens is required — dark objects can still be weighed.

Level 3 — Microlensing fundamentals

Three numbers from a light curve

A simple event is summarized by when it peaked (t_0), how bright it got ($A_{\max}$), and how long it lasted (t_E). Duration depends on lens mass, the distances, and relative speed.

A light curve tells us three basics



Together these describe the lens and its motion.

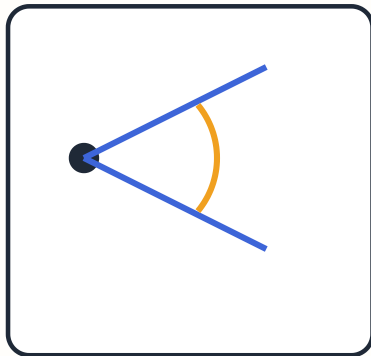
Duration (t_E) depends on mass, distance, and relative speed.

Level 4 — Advanced microlensing

Extra clues reveal the hidden lens

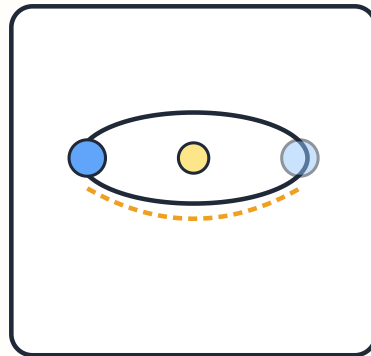
t_E alone cannot give the lens mass. Extra measurables break that degeneracy: Einstein angle, microlensing parallax, and lens light. Usually you need two of these. Without them, a Galactic model supplies statistical estimates.

Extra clues reveal the hidden lens



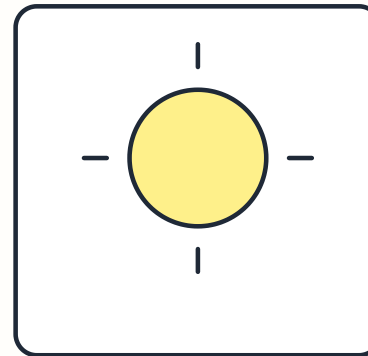
Einstein angle

event size on sky



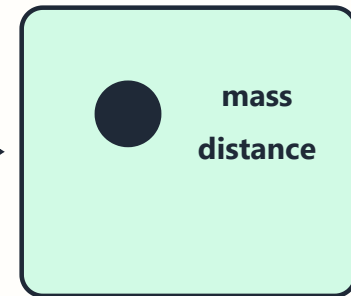
parallax

Earth's viewpoint shift



lens light

host brightness



mass + distance

Usually, two strong clues can break the main degeneracy.

Without them, Galactic models give statistical estimates.

That is the heart of the Hands-On II mass calculation.

Level 4 — Advanced microlensing

Start simple, then add complexity

Begin with one lens. Add a second lens only if the data show a planet or binary. Add parallax or orbital motion only if required. Each new parameter must earn its place.

Start simple, then add complexity



Add a harder model only when the data require it.

This reduces overfitting and false discoveries.

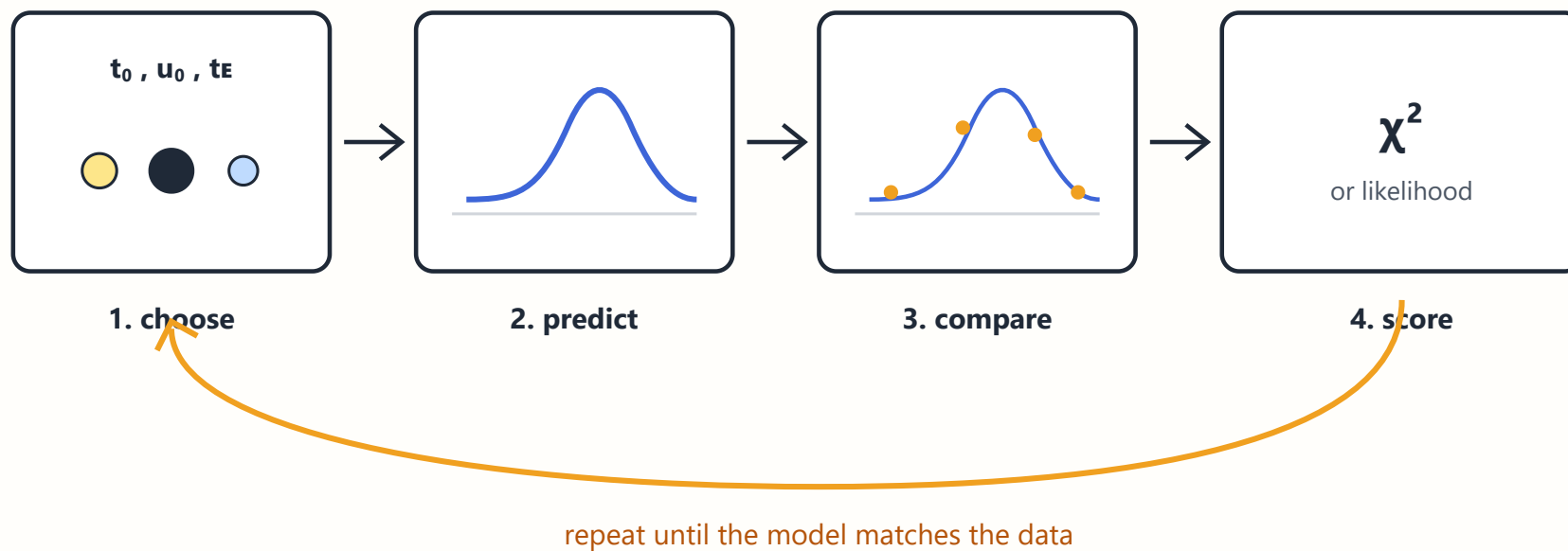
Each new parameter must earn its place in the fit.

Level 4 — Advanced microlensing

Fitting means improving a guess

Every fitter runs the same loop: choose parameters, predict a light curve, compare with the data, score the fit, then choose a better guess. Try many starting points so a good solution is not missed.

Fitting means improving a guess



Try many starting points so a good solution is not missed.

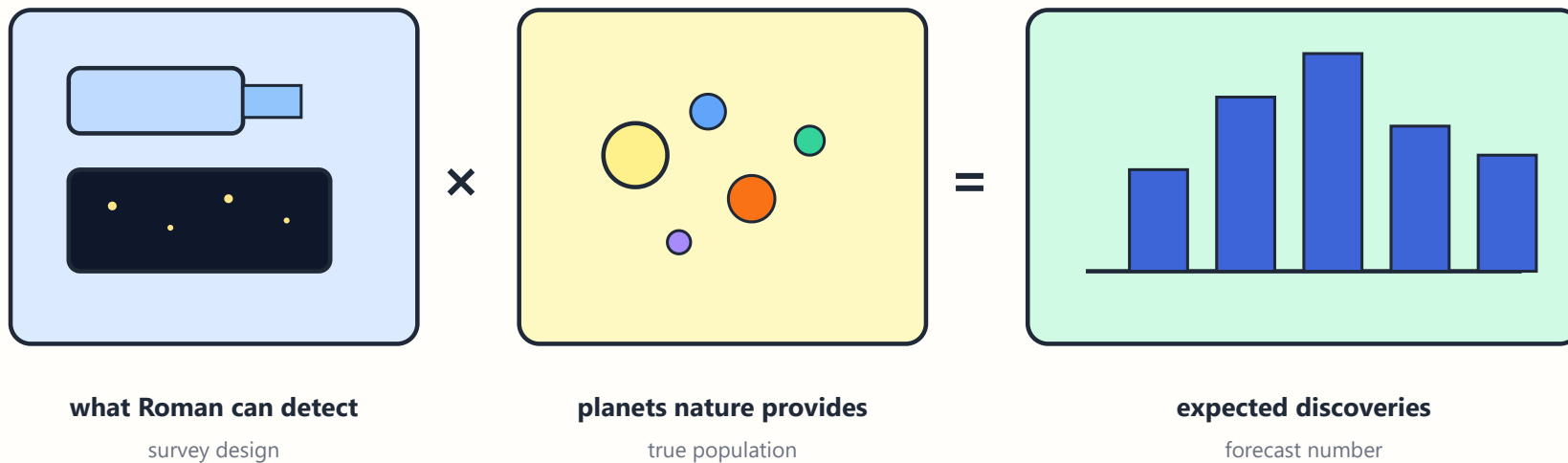
MCMC and nested sampling differ mainly in how the next guess is chosen.

Level 4 — Advanced microlensing

A yield is a prediction

"Roman will find N planets" is a forecast built from what the survey can detect, what planets nature provides, and the resulting expected discoveries. Yield is a forecast — not a promise.

A yield is a prediction



Change the assumptions, and the predicted number changes.

Yield is a forecast — not a promise.

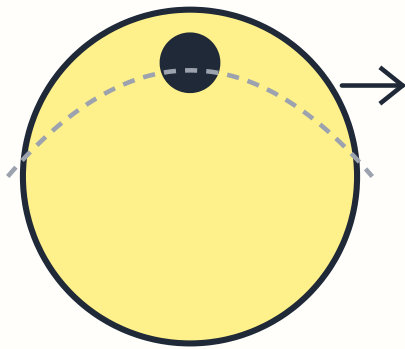
Always ask: which Galaxy model and planet model were used?

Level 5 — Transits

A planet blocks some light

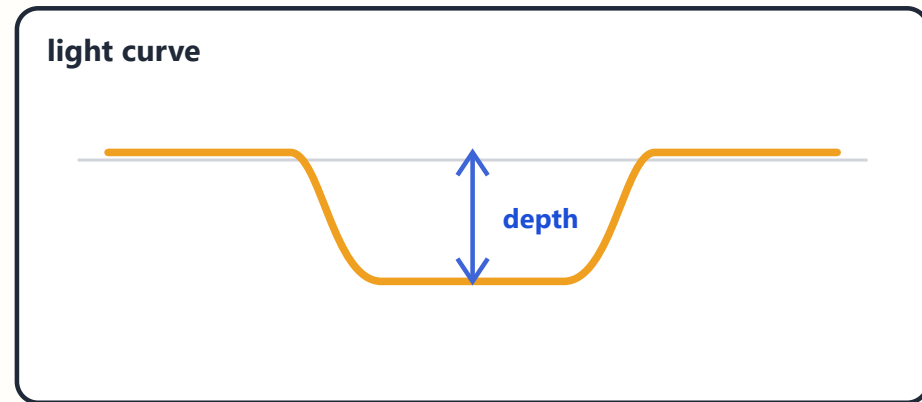
When a planet crosses its star, it blocks a tiny fraction of the light and the light curve shows a small dip. A deeper dip means a larger planet relative to the star. Repeating dips give the orbital period.

Transit = a planet blocks some light



planet crosses star

from our point of view



dip depth → planet size

Deeper dip = bigger planet relative to the star.

Repeating dips give the orbital period.

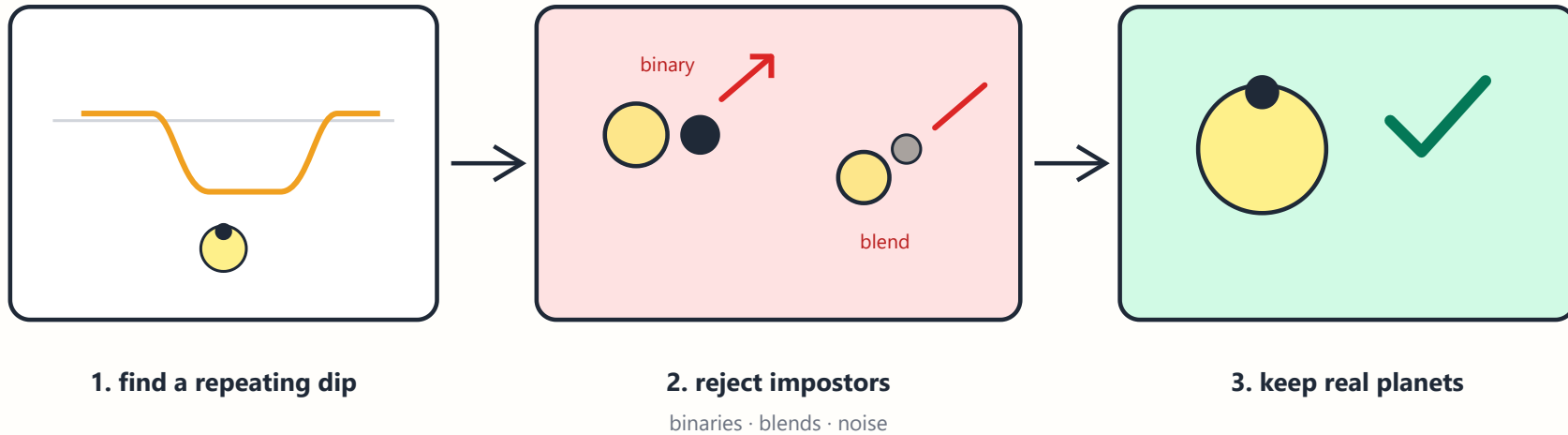
Depth measures planet radius relative to star radius.

Level 5 — Transits

A dip is only a candidate

Noise, eclipsing binaries, and blended neighbors can copy a transit signal. Find a repeating dip, check other causes, and keep only the best survivors.

A dip is only a planet candidate



Noise and binary stars can copy a transit signal.

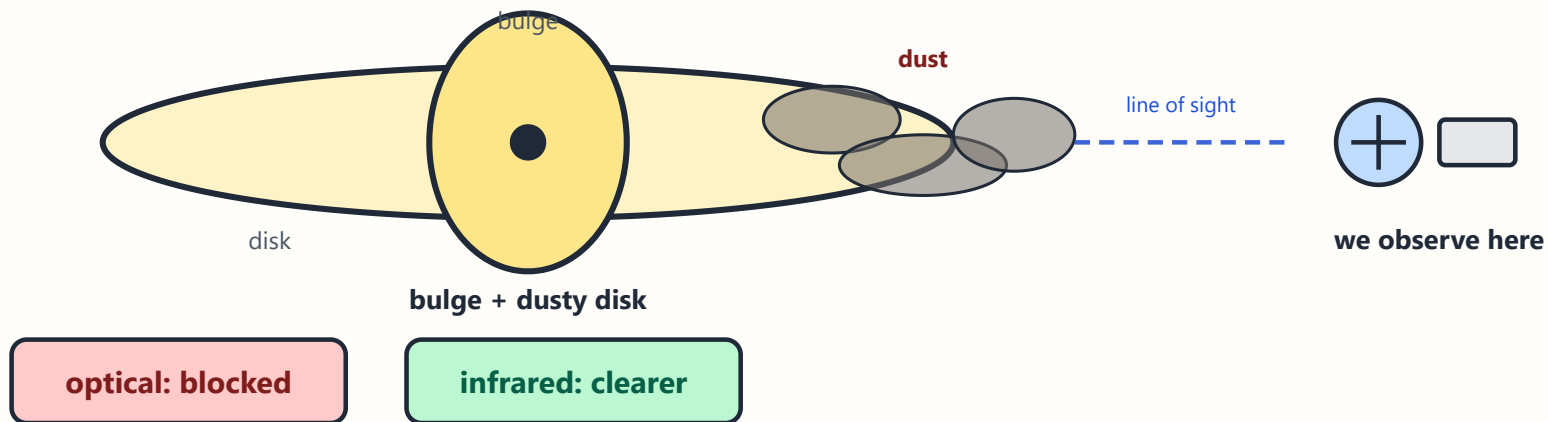
Vetting turns a candidate list into a reliable sample.

Level 6 — Galactic context and dust

The Milky Way changes what we can see

Looking toward the bulge means looking through crowded stars and dust. Infrared helps Roman see through it. Disk, bulge, and nuclear stellar disk overlap — which is why Hands-On II combines CMD, proper motion, and parallax.

The Milky Way changes what we can see



Infrared light helps Roman see through the dust.

Disk, bulge, and nuclear stellar disk overlap toward the center.

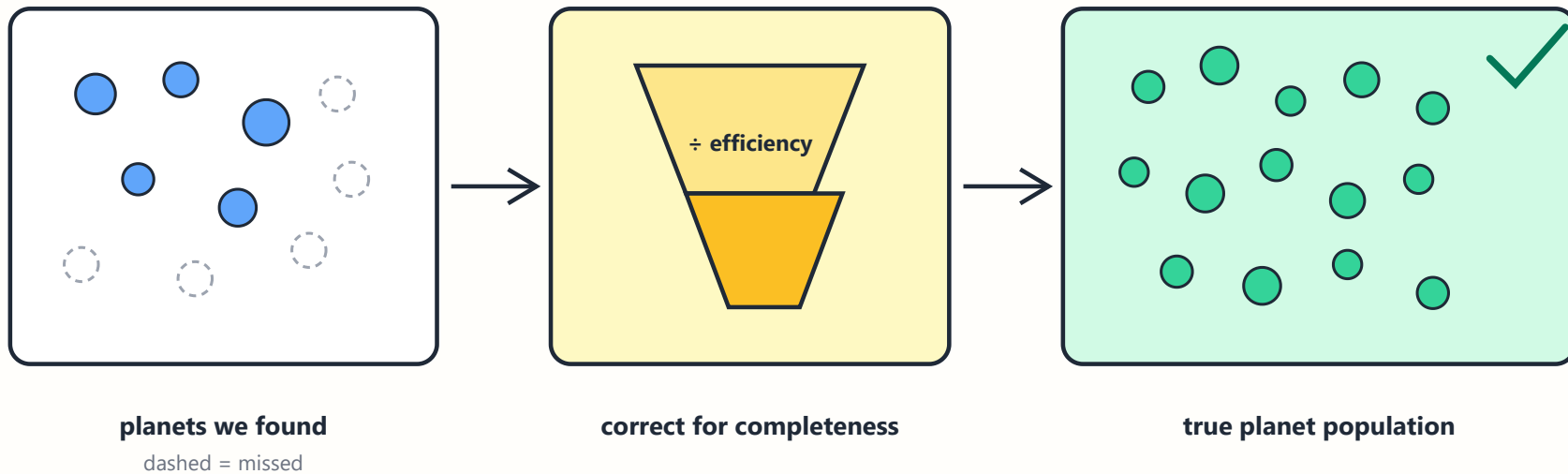
That mix is why Hands-On II needs CMD, proper motion, and parallax.

Level 7 — Occurrence rates

Found planets are only part of the story

Surveys find easy planets and miss hard ones. Correct for the planets that were probably missed — that correction is called completeness. Occurrence rates are detections divided by what could have been found.

Found planets are only part of the story



Hard-to-find planets need a bigger correction.

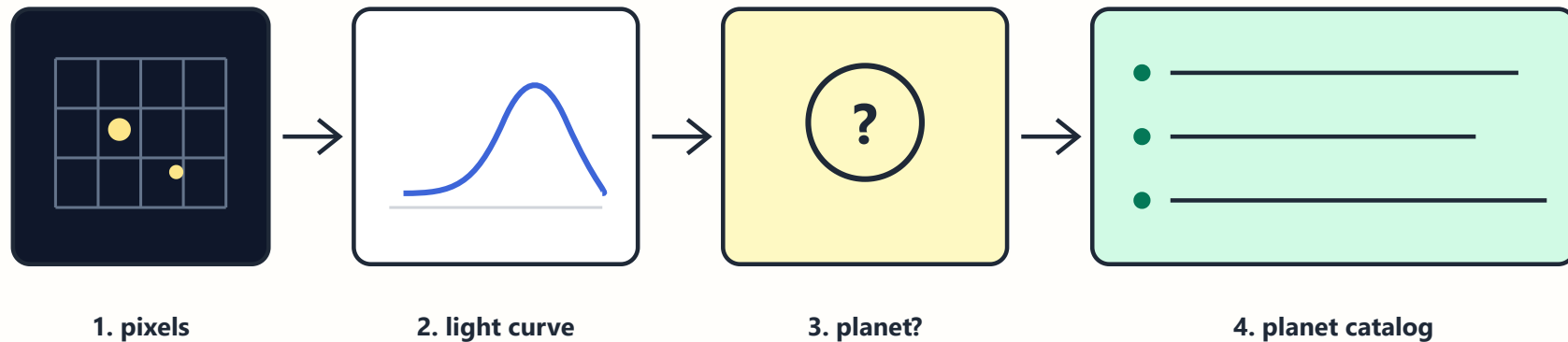
Occurrence rates = detections $\div$ what could have been found.

Level 8 — Pipelines

A pipeline turns pictures into answers

Pixels become brightness-over-time, then a planet-signal check, then a planet list. Fixed steps make results reproducible and selection effects knowable.

A pipeline turns pictures into answers



Each step must be tested, or mistakes move forward.

Fixed steps make results reproducible and selection effects knowable.

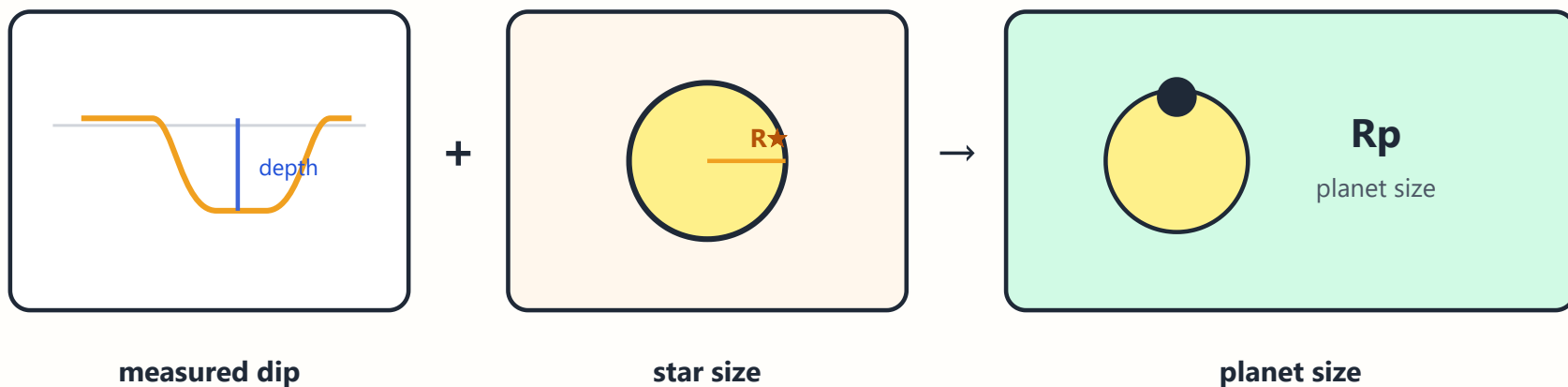
Without that, occurrence rates cannot be trusted.

Level 8 — Host stars

Know the star to know the planet

Transit and microlensing mostly measure ratios between planet and star. Get the star's size or mass wrong and every planet in the sample shifts with it. SED fitting and isochrones pin down the host.

Know the star to know the planet



A wrong star radius gives a wrong planet radius.

Transit and microlensing mostly measure ratios, not absolute sizes.

SED fitting and isochrones are how we pin down the host.

Level 9 — Follow-up

More telescopes add more clues

A Roman detection is often just the start. Earlier images and later images (when lens and source separate) add motion, distance, and mass information the original light curve could not provide.

More telescopes add more clues



Longer timelines help measure motion, distance, and mass.

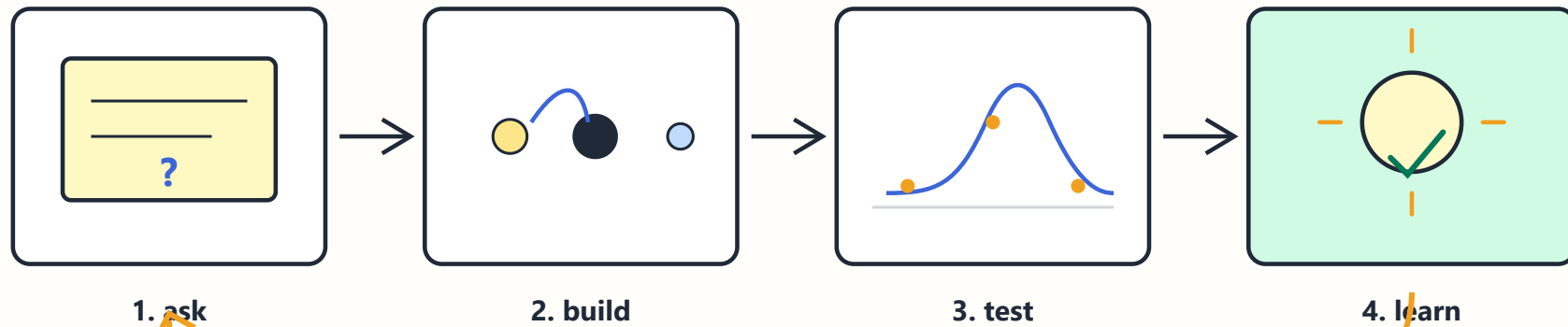
Years later, lens and source can separate enough to be seen apart.

Level 10 — Hands-on roadmap

Science improves in a loop

Ask a question, build a model, test with data, learn, and repeat. The Hands-On II group project notebook is one full lap around this loop.

Science improves in a loop



repeat — a failed prediction shows what to improve

The Hands-On II notebook is one full lap around this loop.

Concepts, data, models, and interpretation feed each other.